Earth which makes the stars turn overhead, and the two methods of reckoning differed by 20 degrees. When the men sailed northward, as reckoned by them, they were actually sailing “north 20 degrees east,” or about 80 percent north and 20 percent east, as reckoned by the women. In this sense, the men’s north was a mixture of the women’s north and east, and similarly the women’s north was a mixture of the men’s north and west.

The key that led Hermann to this discovery was the formula of Pythagoras: Take two legs of a right triangle; square the length of one leg, square the length of the other, add them, and take the square root. The result should be the length of the triangle’s hypotenuse.

The hypotenuse was the straight-line path from Mledina to Serona. The absolute distance along that straight-line path was $\sqrt{210^2 + 100^2}$ = 232.6 furlongs as reckoned using the women’s map with its legs along true east and true north. As reckoned using the men’s map with its legs along magnetic east and magnetic north, the absolute distance was $\sqrt{164.5^2 + 164.5^2}$. furlongs. The eastward distance and the northward distance were “relative”; they depended on whether the map’s reference frame was magnetic or true. But from either pair of relative distances one could compute the same, absolute, straight-line distance.

2.1 The two maps of the route from Mledina to Serona superimposed on each other, together with Hermann’s notations of magnetic north, true north, and the absolute distance.